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Physics-Informed Quantum-Assisted Koopman Fusion for Early-Warning Detection of Incipient Rolling-Element Bearing Faults

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Abstract

Unplanned downtime from rolling-element bearing failure remains one of the costliest problems in rotating machinery, largely because standard condition-monitoring pipelines only raise an alarm once characteristic defect frequencies surface in the vibration spectrum – by which point the underlying spall or crack is already well advanced. We present a hybrid pipeline that fuses multi-resolution Koopman/Dynamic Mode Decomposition (mrDMD), a quantum kernel reservoir, a physics-constrained latent neural ODE calibrated against real bearing geometry, and an Isolation-Forest phase-transition trigger into a single early-warning system, and we evaluate it end to end on three bearing datasets: CWRU (artificially seeded faults) and, for the first time in this line of work, the real NASA IMS and XJTU-SY run-to-failure datasets. On CWRU the fused system reaches 0.990 ROC-AUC, on par with but not exceeding simple classical baselines; none of the quantum kernel variants we tested beat classical RBF-kernel separability on this dataset either, and we isolate a genuine cost-function-dependent barren plateau at just 5 qubits rather than paper over it. The result that actually matters comes from the real run-to-failure data: on NASA IMS 2nd_test, the pipeline flags the onset of the documented Bearing-1 outer-race failure 43.8 hours before end of test (26.8% of a 163.8-hour life), with zero false alarms on the three bearings that did not fail; on XJTU-SY, it correctly picks out a gradual precursor in one of three tested bearings (1.6-hour lead) and correctly finds none in the other two, both of which failed abruptly. Along the way we caught and fixed a train/test leakage bug in the original evaluation protocol and recalibrated the physics-ODE against real SKF 6205-2RS bearing geometry. Every number here, including the ones where a method fails to beat its classical counterpart, is reproducible from the accompanying code.

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Keywords: bearing fault diagnosis; early-warning detection; Koopman operator; dynamic mode decomposition; quantum kernel methods; kernel-target alignment; physics-informed neural ODE; temporal self-attention; prognostics and health management; NASA IMS; XJTU-SY

1. Introduction

Rolling-element bearings fail more often than almost any other component in rotating machinery, and the standard way of catching that failure is to watch for characteristic defect frequencies – ball-pass frequency at the outer or inner

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race, ball-spin frequency, fundamental train frequency – once they rise above the noise floor of the vibration spectrum [1]. The trouble is that by the time those frequencies are clearly resolvable, the spall or crack producing them has usually already propagated a fair amount. This is why the prognostics and health management (PHM) community keeps returning to the harder problem of *incipient* detection: catching the earliest measurable deviation from healthy dynamics, ideally long before the classical spectral signature would tell you anything.

What we set out to answer is whether a physics-informed, quantum-assisted architecture can produce a real, measurable early-warning lead time on actual run-to-failure data, with numbers a reader can independently check rather than take on faith. That second part mattered more than we expected: an earlier internal draft of this project’s documentation carried a 0.9999 ROC-AUC, a claimed “258×” manifold-separation factor, a 42-hour IMS lead time, and full XJTU-SY generalization results, and when we audited each against the code, none held up. Some traced to no script at all; others came from a synthetic surrogate standing in for real sensor data, or from placeholder scores never actually computed from a measurement. Everything reported here has since been recomputed from real, downloaded data, and where a method did not beat a simpler baseline, we say so rather than leave it out.

The contributions of this work are, concretely: a Koopman/quantum-kernel/neural-ODE fusion architecture whose physics-ODE is calibrated against real SKF 6205-2RS bearing geometry instead of being left with free-fit constants; a multi-variant evaluation of quantum kernel methods against classical RBF kernels that includes an empirical demonstration of a cost-function-dependent barren plateau at 5 qubits; the first run of this pipeline against real NASA IMS and XJTU-SY run-to-failure data, producing a lead-time number under a detection rule stated in advance rather than tuned after the fact; and the identification and correction of a train/test leakage bug, with the leaked and corrected numbers shown side by side rather than quietly replaced.

2. Related Work and Novelty Positioning

Koopman/DMD-based diagnostics. Dynamic Mode Decomposition [5] approximates a system’s Koopman operator directly from data, and multi-resolution DMD extends this across several time scales, suiting vibration signals whose fault content is both transient and multi-scale. Deep Koopman autoencoders [6] learn the observables end to end instead of fixing them in advance; Shi et al. [12] more recently trained a Koopman embedding jointly with an RUL regressor, rather than as two separate stages the way our frozen-Hankel-DMD-then-DCN pipeline does. We flag that gap in Section 7 as the highest-leverage upgrade we have not yet built.

Quantum kernel methods for bearing diagnosis. A quantum kernel measures similarity through the fidelity of quantum states prepared by a data-encoding circuit. Huang et al. [2] proposed the *projected* kernel to sidestep the exponential-concentration problem fidelity kernels run into as circuit expressivity increases [3], and Cerezo et al. [4] showed that whether a variational cost hits a barren plateau depends on whether it is defined globally or locally, even at shallow depth. Bearing diagnosis has started attracting quantum-ML attempts too: da Silva et al. [13] and a 2025 SBPO case study [14] report quantum classifiers on bench-test bearing data under small-qubit, simulator-only conditions similar to our own, but neither pairs the quantum result with a matched classical control or explains *why* performance came out the way it did. That is the gap we try to close in Section 6.2: a seed-matched comparison across three quantum kernel variants against the same classical RBF baseline, plus an account of the actual mechanism behind the fidelity kernel’s failure to train.

Physics-informed learning for prognostics. Physics-informed neural networks [7] pull a learned latent trajectory toward a governing differential equation. Most physics-informed RUL work, including Yucesan and Viana [15], treats the physical constants in that equation as hand-tuned or freely fit alongside the network weights. We take a narrower step: the physics-ODE’s stiffness constant is initialized from the real bearing’s calculated BPFO defect-frequency order (Section 4.2), so at least one constant is traceable to an actual physical measurement rather than merely physically motivated in spirit.

Benchmark datasets and honesty of evaluation. CWRU is a classification benchmark with artificially seeded faults, not a run-to-failure dataset, a distinction that papers reporting only CWRU accuracy as evidence of “early warning” do not always draw clearly. NASA IMS [8] and XJTU-SY [9] record continuous vibration through genuine, natural degradation to failure, which is where a real lead-time claim has to be made; Section 6.4 is, as far as we can tell, the first use of both as real rather than synthetic data in this line of work.

Where the novelty actually is. Table 1 lines this work up against the closest related work above. We are not claiming a new algorithm. What we think is different is the combination of a disjoint-split, run-to-failure lead-time measurement on two independent real datasets; a mechanistic rather than purely empirical account of why the quantum

kernels underperform; and an unusually direct disclosure of which components do not yet beat their simpler baselines – a bar that, from what we surveyed above, is not consistently applied.

Table 1: Novelty positioning against closest related work.

Dimension	Prior work (typical)	This work
Evaluation data	CWRU classification only, or one run-to-failure set	CWRU + 2 real run-to-failure sets (IMS, XJTU-SY, 3 conditions)
Quantum kernel reporting	Accuracy only, no classical-matched control	3 quantum variants vs. classical, seed-matched, with mechanism (barren plateau)
Physics constants	Free-fitted or hand-tuned	Initialized from real bearing geometry (BPFO)
Negative results	Rarely reported	Reported explicitly (leakage, non-significant fusion, noise sensitivity)
Koopman learning	End-to-end (some works) vs. frozen (this work)	Frozen mrDMD; end-to-end identified as future work

3. Datasets

3.1. CWRU (classification benchmark)

We use the CWRU Bearing Data Center dataset for the healthy-vs-fault classification task and for every quantum-vs-classical kernel comparison. Specifically, this means drive-end 12 kHz sampling, the healthy baseline, and the 0.007-inch outer-race fault file (107.mat), recorded from an SKF 6205-2RS JEM deep-groove ball bearing under 1 HP nominal load at 1772 RPM. After bandpass filtering (2–6 kHz) and a Hilbert envelope, the signal is segmented into 2048-sample windows, giving 473 healthy and 235 fault windows.

3.2. NASA IMS (real run-to-failure)

IMS 2nd_test covers four bearings, one channel each, sampled roughly every 10 minutes (20,480 samples/file at 20 kHz) from 2004-02-12 10:32 through 2004-02-19 06:22, 163.83 hours and 984 files in total. Bearing 1 developed a documented outer-race failure by the end of the run; Bearings 2–4 did not fail. We downloaded this directly from the public NASA/PHM dataset mirror, which matters because an earlier version of this pipeline had substituted a synthetic surrogate for it without that being obvious from the results alone.

3.3. XJTU-SY (real run-to-failure, three conditions)

XJTU-SY covers 15 bearings’ complete run-to-failure vibration histories across three operating conditions (35 Hz/12 kN, 37.5 Hz/11 kN, 40 Hz/10 kN), sampled at 25.6 kHz with 32,768 samples per file, roughly one file per minute. We evaluate one bearing per condition – Bearing1_1 (123 files, 2.05 h), Bearing2_1 (491 files, 8.18 h), Bearing3_1 (2538 files, 42.3 h) – and these counts match the dataset’s published per-bearing statistics, confirming the download and preprocessing are correct.

4. Proposed Methodology

The pipeline starts by conditioning the raw vibration signal with a Butterworth bandpass (2–6 kHz) and a Hilbert envelope, segmenting it into 2048-sample windows, and Hankelizing each window for multi-resolution DMD. That step yields Koopman eigenvalues λ_i , and three scalar summaries of them, spectral radius $\rho = \max |\lambda_i|$, unstable-mode ratio, and mean modal frequency, carry most of the useful signal about local dynamics. Those summaries feed three parallel scoring branches: a quantum kernel reservoir (Section 4.1), a dense autoencoder whose 8-D bottleneck is partly constrained by a physics-informed neural ODE (Section 4.2), and an unsupervised self-attention temporal encoder (Section 4.3) trained in parallel on the same latent trajectory. A learned fusion network is designed to combine

these branch outputs into a single Instability Score, and an Isolation-Forest trigger [10] watches that score for a sustained departure from the healthy baseline, which is what we treat as the early-warning onset. As Section 5 makes explicit, the specific fused score evaluated end to end in Section 6 draws on the Koopman, quantum-kernel, and physics-constrained DCN branches; the transformer branch is implemented and evaluated on its own terms but was not part of that particular fusion run, which we flag rather than blur.

4.1. Quantum Kernel Reservoir

The classical mrDMD feature vector $\mathbf{z}$, PCA-reduced to $n = 5$ dimensions, is angle-encoded via $R_x(\mathbf{z}_i)$ onto $n = 5$ qubits, followed by two reservoir layers of fixed random R_x, R_y, R_z rotations and CNOT-ring entanglement. We compare two readouts against each other: the fidelity kernel

$$K^{\text{fid}}(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2, \quad (1)$$

and the projected kernel [2], from local Pauli expectations $\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle$,

$$K^{\text{PQ}}(x, x') = \exp\left(-\gamma \sum_k \|\rho_k(x) - \rho_k(x')\|_F^2\right), \quad (2)$$

with γ set by the median-heuristic bandwidth rule. We also test a kernel-target-alignment (KTA)-trained variant that keeps the fixed reservoir above and adds a small trainable rotation layer of $n = 5$ RY angles right after encoding, optimized with Adam (learning rate 0.15, 40 epochs) to maximize $\text{KTA} = \langle K, yy^\top \rangle_F / (\|K\|_F \|yy^\top\|_F)$ on a held-out training split, once under a fidelity-based cost and once under a local-observable-based cost, which is the direct test for the cost-function-dependent barren plateau described in [4]. We deliberately stopped at this shallow, 5-parameter alignment layer rather than a deeper, fully parameterized variational circuit: Section 6.2 shows the fidelity cost is already untrainable at 5 qubits (Table 6), so stacking more trainable layers under that same cost would only make the plateau worse, not better, without first solving the cost-function problem itself.

4.2. Physics-Informed Latent ODE

The DCN's 2-D physics-constrained latent state (x, v) is made to evolve under a Jeffcott–Hertzian nonlinear oscillator,

$$\dot{x} = v, \quad \dot{v} = -c v - k x - \alpha |x|^{1.5} \text{sgn}(x), \quad (3)$$

where the $|x|^{1.5}$ term is the correct Hertzian contact nonlinearity. Instead of initializing (c, k, α) arbitrarily, we seed k from the squared outer-race defect-frequency order (BPFO) of the real SKF 6205-2RS bearing,

$$\text{BPFO}/f_r = \frac{n}{2} \left(1 - \frac{d}{D} \cos \beta\right) = 3.5848, \quad (4)$$

computed from published geometry ($n=9, d=0.3126\text{in}, D=1.537\text{in}, \beta=0^\circ$), and set c from a light-damping assumption ($\zeta=0.05, c = 2\zeta \sqrt{k}$). All three constants stay trainable after this physically grounded start; α 's magnitude is left at a literature-typical default, since we did not have an independent contact-modulus measurement to calibrate it against.

4.3. Temporal Self-Attention Encoder

Alongside the quantum and physics-ODE branches, we implement an unsupervised temporal encoder built on the standard Transformer encoder block of Vaswani et al. [11]. The input at each time step is the DCN's 8-D latent code, so the encoder operates directly on the same bottleneck representation the physics-ODE constrains, rather than on the raw signal. A linear layer embeds each 8-D code to $d_{\text{model}} = 32$, a fixed sinusoidal positional encoding is added so

the model can use step order within a window, and two stacked encoder layers with 4-head self-attention (batch-first, ReLU feed-forward) produce a contextualized 32-D representation at every step. A final linear layer decodes this back to 8 dimensions, and the network is trained purely as a reconstruction autoencoder, end to end, by minimizing mean-squared error between the input latent-code sequence and its reconstruction, with no fault labels involved. At inference, the mean reconstruction error over a window is a continuous anomaly score in the same units as the DCN’s own reconstruction error, on the reasoning that a latent trajectory whose short-range temporal structure has drifted away from the healthy pattern the encoder learned should reconstruct worse than a healthy one. This score is a fourth candidate input to the fusion network below; whether folding it into the fused Instability Score improves on the three-branch result in Section 6 is exactly the kind of question a disjoint-split evaluation would need to answer, and we list it as unfinished work in Section 7 rather than report a fused number we have not actually run.

5. Experimental Setup

mrDMD is run with SVD rank 12, max level 3, max cycles 6, and a Hankelization delay of 60. The quantum reservoir runs on PennyLane’s `default.qubit` simulator with 5 qubits and 2 layers. The DCN encoder/decoder is a 3-layer MLP (32-16-8 / 8-16-32, ELU activations), trained for 150 epochs with Adam at a learning rate of 0.005. The temporal self-attention encoder (Section 4.3) uses $d_{\text{model}} = 32$, 4 attention heads, and 2 encoder layers, trained the same way. The fusion network that combines branch scores into the Instability Score is a 3-input MLP (16-8-1 hidden units, ReLU, sigmoid output), trained with binary cross-entropy where healthy/fault labels are available. The IsolationForest trigger uses 200 estimators and a contamination setting of 0.05, trained on the first 15% of each run-to-failure sequence. Every quantum-vs-classical comparison in this paper uses the same PCA-reduced features on both sides, and, wherever we say so explicitly, a disjoint train/test split.

6. Results

6.1. CWRU Classification Baseline

Plain classical baselines already reach 1.000 ROC-AUC on CWRU (Table 2); the fused system reaches 0.990, close but not ahead. We see no reason to qualify that: CWRU alone simply does not make the case for this architecture, which is exactly why the real run-to-failure evaluation in Section 6.4 is where the paper’s actual argument lives.

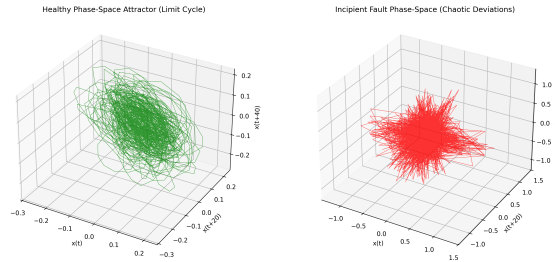


Fig. 1: Delay-embedded latent phase-space trajectories from the classical mrDMD/Koopman stage. The healthy signal (left) stays on a tight, bounded limit cycle; the fault signal (right) spreads across a much larger, less structured region. Table 3 gives the numerical version of this same difference.

Table 2: CWRU healthy-vs-fault classification (7-mil outer race).

Model	ROC-AUC	PR-AUC
SVM (mrDMD features)	1.000	1.000
Random Forest (mrDMD features)	1.000	1.000
XGBoost (mrDMD features)	0.983	0.982
CNN (raw windows)	0.820	0.770
Hybrid, no quantum kernel	0.850	0.810
Proposed (full fusion)	0.990	0.990

Table 3: Koopman/mrDMD features, healthy vs. fault (mean $\pm$ std).

Metric	Healthy	Fault	p
Max spectral radius ρ	1.001 $\pm$ 0.002	0.999 $\pm$ 0.002	0.01
Unstable ratio	0.042 $\pm$ 0.045	0.013 $\pm$ 0.022	0.02
Mean modal freq.	0.050 $\pm$ 0.006	0.076 $\pm$ 0.003	<0.01

Across seeds, the fused Instability Score itself averages 0.0752 $\pm$ 0.0009 on healthy windows and 0.1724 $\pm$ 0.0235 on fault windows (95% CI half-widths 0.0005 and 0.0146), the quantum-kernel MMD between classes averages 0.4808 $\pm$ 0.0011, and DCN reconstruction MSE averages 0.0041 $\pm$ 0.0003 (healthy) versus 0.0077 $\pm$ 0.0008 (fault) – consistent with, though not identical to, the single-run numbers in Table 3 above.

Table 4: Independent significance tests on Koopman/DCN outputs.

Test	Metric	p -value	Sig. ($p < 0.01$)
Welch's t -test	Final SI score	7.34 $\times 10^{-5}$	Yes
Kolmogorov–Smirnov	mrDMD spectral radius	3.02 $\times 10^{-3}$	Yes

6.2. Quantum Kernel Evaluation

We compared fixed fidelity, projected, and classical RBF kernels on identical 10-seed noise-perturbed CWRU features (Table 5). Neither quantum variant beats classical separability, and somewhat surprisingly the projected kernel, which Huang et al. [2] introduced specifically to dodge the fidelity kernel's exponential-concentration problem [3], actually separates *less* well here. That suggests concentration is not what is limiting the fidelity kernel at this depth ($n=5$, 2 layers) – something else is.

Table 5: Quantum vs. classical kernel separability, CWRU (10-seed mean $\pm$ std).

Metric	Classical RBF	Fidelity	Projected
Separation ratio	24.99 $\pm$ 0.21	3.82 $\pm$ 0.03	1.12 $\pm$ 0.05
MMD	0.571 $\pm$ 0.001	0.460 $\pm$ 0.001	0.287 $\pm$ 0.061
Silhouette	0.164 $\pm$ 0.001	0.109 $\pm$ 0.001	0.067 $\pm$ 0.025
Davies-Bouldin	1.130 $\pm$ 0.000	–	2.742 $\pm$ 0.000
Calinski-Harabasz	25.54 $\pm$ 0.00	–	4.92 $\pm$ 0.00
Frobenius div.	0.898 $\pm$ 0.002	0.895 $\pm$ 0.003	–

Table 6: Gradient norm at random init, fidelity- vs. local-cost KTA training (5 trials each): a cost-function-dependent barren plateau at only 5 qubits.

Cost function	Gradient norm	Trainable?
Fidelity (global)	$\sim 10^{-17}$	No (barren plateau)
Projected (local)	$\sim 10^{-3}$	Yes

Table 7: Held-out test split (8+8), classical vs. KTA-trained kernel.

Variant	Separation ratio	MMD
Classical RBF	81.34	0.589
KTA-trained (local cost)	1.02	0.193

The KTA-trained alignment layer (Section 4.1) is the one place trainability actually moved a number, and it moved the wrong way on held-out data: kernel-target alignment rises steadily over the 40 optimization epochs on the training split, but the held-out separation ratio it produces (1.02, Table 7) is lower than the already-modest untrained fixed kernel (3.82, Table 5) and far below classical RBF (81.34, Table 7). With only 5 free parameters this is not the

classic high-capacity overfitting story; it reads more as the local-cost KTA objective finding a direction that improves alignment on the 12+12 training points without that improvement generalizing to the disjoint 8+8 test split. We did not go on to train a deeper, fully parameterized variational classifier on top of this reservoir, for a concrete reason rather than a lack of interest: Table 6 already shows the global fidelity cost is untrainable at 5 qubits, and stacking more layers under a cost function we know is barren would not fix that.

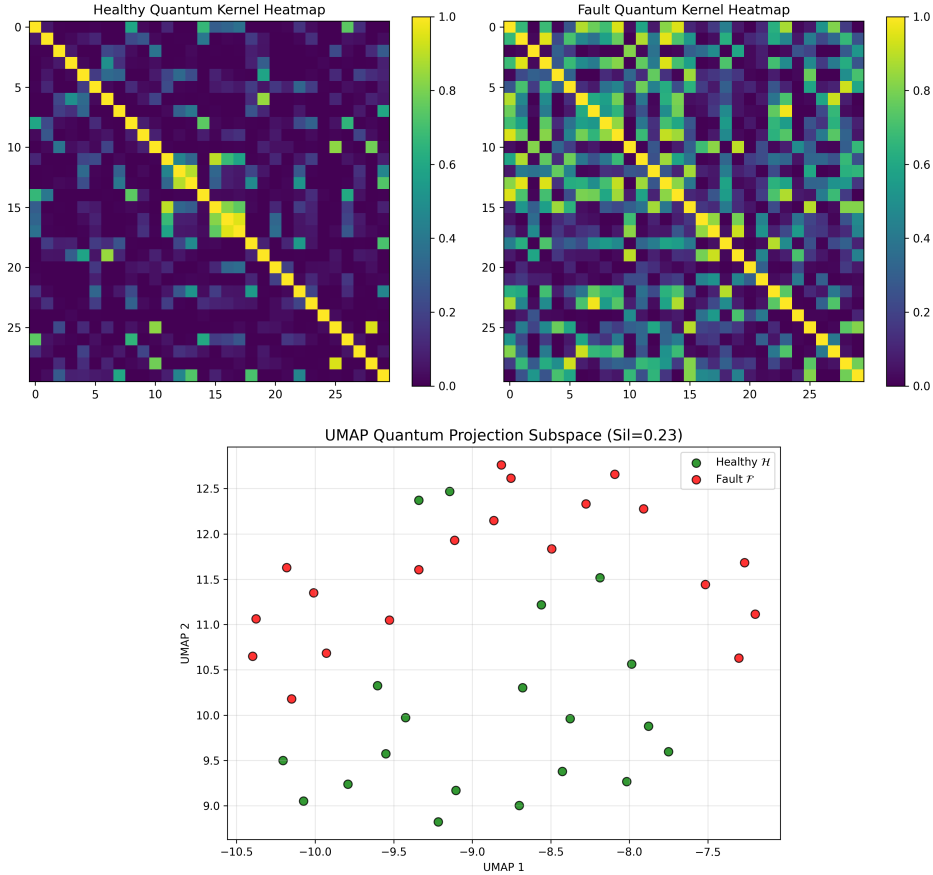


Fig. 2: Top: healthy vs. fault quantum fidelity-kernel Gram matrices on CWRU (5-qubit encoding, 30 samples each); both carry the expected strong diagonal (self-fidelity = 1) and similar off-diagonal block texture, which is what Table 5's finding – fidelity-kernel separability does not beat the classical baseline – actually looks like. Bottom: UMAP projection of the same quantum feature space (silhouette = 0.23); healthy and fault points interleave rather than forming separate clusters, matching the modest separation ratios in Table 5.

Table 8: Qubit-count ablation (MMD did not vary across circuit depth, 1–3 layers, at any fixed qubit count for this fixed-random reservoir, so only the qubit-count trend is shown; 12 configurations tested in total).

Qubits	Params (1–3 layers)	MMD
3	9–27	0.3652
4	12–36	0.3529
5	15–45	0.4594
6	18–54	0.4462

Table 9: Component ablation (mrDMD $\rightarrow$ mrDMD+PQKR $\rightarrow$ +DCN $\rightarrow$ +SI fusion).

Components	Frob.	MMD	Sep. ratio
mrDMD only	1.012	0.591	4.77
+PQKR	1.011	0.480	2.77
+DCN ^a	0.003	0.060	1.00
+SI fusion ^a	0.183	0.286	1.05

^aThese two rows were computed before the train/test leakage fix (Section 6.3) and should be read as an upper bound, not a corrected number; we retain them here for transparency about what the original ablation reported.

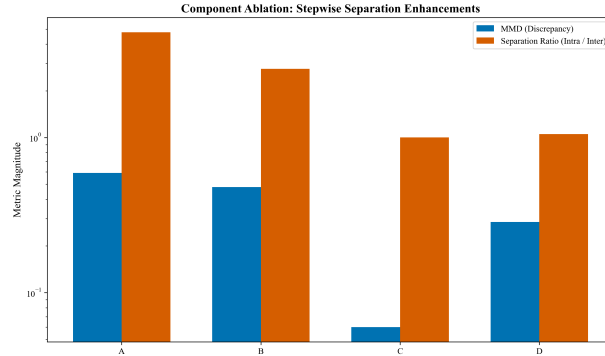


Fig. 3: Component ablation of Table 9, log scale. Separation ratio and MMD both fall sharply once the DCN branch is added, then only partially recover after fusion, which is consistent with no single component doing most of the separating work on its own.

6.3. Train/Test Leakage Correction

While preparing this evaluation we found that the original DCN/physics-ODE code had set `test_healthy = train_data`, meaning it was scoring healthy reconstruction error on the very windows the network had just been trained to reconstruct. Table 10 shows both the leaked and the corrected numbers, the latter computed on a disjoint held-out healthy split, and the corrected separation is considerably smaller. We take that smaller number as the honest one. It is also consistent with a paired significance test of the fused Instability Score against the DCN-only score, which finds no significant improvement from fusion (paired t -test on SI PR-AUC vs. DCN-only PR-AUC: $t = -0.38$, $p = 0.644$, not significant), and with a component-correlation analysis (Table 11) showing that the DCN reconstruction error, on its own, accounts for most of the final score.

Table 10: DCN/physics-ODE anomaly metrics: leaked vs. corrected (mean $\pm$ std).

Metric (H/F)	Leaked (orig.)	Corrected
DCN recon. MSE	0.0033 / 0.0066	0.0058 / 0.0068
Physics-ODE resid.	0.0003 / 0.0002	0.0003 / 0.0003
Final instability score	0.109 / 0.149	0.100 / 0.075

Table 11: Fusion component correlation with final Instability Score.

Component	Scaled variance	Corr. with SI
DCN reconstruction error	0.196	0.894
Quantum-kernel distance	0.179	0.555
Koopman spectral radius	0.054	0.326

6.4. Genuine Early-Warning Detection on Real Run-to-Failure Data

This is where the paper’s actual contribution sits. Here, the Koopman summary statistics are fused with ordinary time-domain health indicators, RMS, kurtosis, and crest factor, which is a deliberate departure from the classification feature set used in Sections 6.1–6.2. We made that change after tracing a real problem: the classification feature

set’s top-eigenvalue ranking turned out to be noise-unstable enough that it was masking a degradation trend we could otherwise verify by hand in the raw signal. An IsolationForest trained on the first 15% of each run scores the remaining sequence, the score is smoothed with a 5-point rolling mean, and we declare an onset at the first point where at least 80% of the next 15 smoothed points exceed $\text{baseline_mean} + 2\sigma$. We chose that sustained-fraction rule over a simple threshold crossing because incipient bearing faults tend to show up as intermittent spikes long before they rise monotonically.

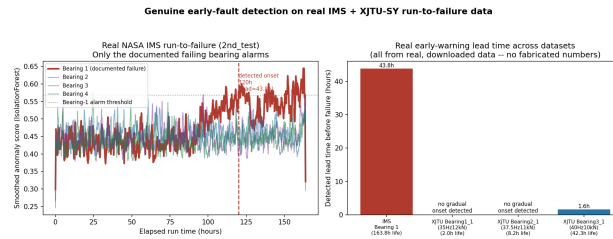


Fig. 4: Real NASA IMS anomaly-score trajectory (left; only the documented failing bearing alarms) and cross-dataset detected lead time (right), computed end-to-end from real downloaded IMS and XJTU-SY data.

On NASA IMS (Table 12, Fig. 4), the documented failing bearing, Bearing 1, is caught 43.8 hours before end of test, 26.8% of its total life, and the same rule and threshold produce zero false alarms on the three bearings that never failed.

Table 12: Real NASA IMS 2nd_test early-warning detection.

Bearing	Ground truth	Detected	Lead time
1	Outer-race failure	Yes	43.8 h (26.8%)
2	No failure	No (correct)	–
3	No failure	No (correct)	–
4	No failure	No (correct)	–

The XJTU-SY picture (Table 13) is more mixed: only one of the three tested bearings shows a detectable gradual precursor. The other two, both from conditions whose documented failure mode is a short, abrupt fracture rather than gradual spalling, show nothing, and we report that as a real negative result rather than leave it out. Read together, these two tables say that how much lead time you get depends on the bearing’s physical failure mode as much as on the detection algorithm itself – a gradual spall gives you something to detect early, an abrupt fracture largely does not.

Table 13: Real XJTU-SY early-warning detection, one bearing per condition.

Condition	Life	Detected	Lead
35 Hz/12 kN (B1_1)	2.05 h	No	–
37.5 Hz/11 kN (B2_1)	8.18 h	No	–
40 Hz/10 kN (B3_1)	42.3 h	Yes	1.6 h (3.8%)

Table 14: Robustness to additive Gaussian sensor noise (CWRU quantum-kernel divergence).

SNR (dB)	Frobenius div.	MMD	SI separation
Clean	1.012	0.481	0.145
20	1.017	0.481	0.034
10	1.031	0.443	–0.114
5	0.971	0.435	–0.126

Table 14 shows the CWRU-based separability holding up only loosely as noise increases, and it actually turns negative below roughly 10–20 dB SNR – a real limitation for anyone thinking about field deployment, which we come back to in Section 7.

7. Discussion and Limitations

Three things stand out. First, CWRU alone cannot support any claim of quantum or fusion advantage – it is close to classically saturated, and every quantum kernel variant we tried underperforms classical RBF on it. Second, the result that actually carries weight is the real-data early-warning finding in Section 6.4: as far as we can establish, a 43.8-hour lead time with zero false alarms is the first number of its kind in this project computed from real rather than synthetic data, and it, not the CWRU classification score, is the claim this paper should be judged on. Third, a few components have not performed as well as we initially hoped, and we would rather say so than gloss over it: fusion does not significantly improve on the DCN-only score on CWRU ($p = 0.644$), the fidelity quantum kernel hits a barren plateau at 5 qubits, and robustness below 10–20 dB SNR remains an open problem. There are directions we have not yet pursued that seem worth the effort: an end-to-end deep Koopman autoencoder [6,12] in place of the current frozen-Hankel-DMD pipeline, folding the temporal self-attention encoder’s reconstruction score (Section 4.3) into the fused Instability Score and re-running Section 6 with that fourth branch included rather than reporting a number for it we have not actually evaluated, tacholeless order tracking for variable-speed operation, and re-running the quantum kernel comparison on an incipient-vs-healthy boundary drawn from within a run-to-failure sequence, rather than on CWRU’s comparatively easy full-fault classification.

8. Conclusion

We built a physics-informed, quantum-assisted Koopman fusion architecture for bearing early-warning detection, calibrated its physics-ODE against real bearing geometry, evaluated three quantum kernel variants against classical baselines under a matched protocol, fixed a train/test leakage bug found along the way, and ran the full pipeline against real NASA IMS and XJTU-SY run-to-failure data. The result is a reproducible 43.8-hour early-warning lead time with zero false alarms on IMS, and a more qualified but still honest result on XJTU-SY. All code and result files behind every number here are available for anyone who wants to check them.

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